

ESENGULY, Turkmenistan

WMO: 387500

Lat: 37.467N

Lon: 53.967E

Elev: -22

StdP: 101.59

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-0.8	-6.2	2.2	1.4	-4.1	2.7	2.9	8.3	8.9	7.4	9.1	2.1	20	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	34.2	24.5	33.0	24.9	32.1	24.7	27.9	31.3	27.2	30.9	26.5	30.5	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.3	26.1	21.4	30.1	25.3	20.4	29.7	89.2	31.4	85.7	31.0	82.8	30.6	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.7	6.7	-5.3	38.9	2.9	2.7	-7.3	40.9	-9.0	42.4	-10.6	43.9	-12.7	45.9
			-6.0	28.9	2.6	1.4	-7.9	29.9	-9.5	30.7	-10.9	31.5	-12.9	32.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.7	7.5	11.2	15.8	21.4	25.6	28.3	28.9	25.1	19.2	12.5	8.2	(6)
(7)		DBStd	8.48	3.19	3.44	3.34	3.41	2.94	2.27	1.80	1.66	2.52	3.33	3.77	3.17	(7)
(8)		HDD10.0	303	109	80	22	1	0	0	0	0	0	16	74		(8)
(9)		HDD18.3	1506	361	303	224	92	6	0	0	0	30	176	314		(9)
(10)		CDD10.0	3075	7	11	59	174	355	469	566	584	454	286	92	18	(10)
(11)		CDD18.3	1237	0	0	2	15	103	219	308	326	204	57	2	0	(11)
(12)		CDH23.3	13146	0	3	44	157	659	2007	3672	4209	2008	363	22	2	(12)
(13)		CDH26.7	4979	0	0	14	56	203	610	1456	1931	630	77	4	0	(13)
(14)	Wind	WSAvg	2.9	2.3	2.6	3.0	3.3	3.4	3.6	3.1	2.7	2.4	2.1	2.1	(14)	
(15)	Precipitation	PrecAvg	207	23	20	28	19	13	5	7	8	13	19	26	26	(15)
(16)		PrecMax	320	74	58	104	53	52	34	37	46	57	92	69	58	(16)
(17)		PrecMin	84	2	2	0	0	0	0	0	0	0	0	1		(17)
(18)		PrecStd	58	17	12	18	16	14	9	11	11	15	22	18	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.3	28.2	32.1	35.0	35.2	35.4	35.8	33.8	32.3	26.0	22.0	(19)
(20)		0.4%	MCWB	11.7	12.6	15.1	17.5	19.8	21.8	25.4	25.2	23.3	18.9	16.8	13.5	(20)
(21)		2%	DB	15.9	17.7	23.3	27.9	30.9	32.2	33.5	34.2	32.0	28.6	22.5	17.5	(21)
(22)		2%	MCWB	10.3	11.5	13.9	16.9	19.8	23.6	25.7	25.7	23.5	19.3	16.2	12.0	(22)
(23)		5%	DB	13.3	14.8	19.8	24.5	28.6	30.9	32.5	33.1	30.9	26.7	20.4	15.1	(23)
(24)		5%	MCWB	9.6	9.9	13.2	16.1	19.7	23.5	25.5	25.8	23.3	19.5	15.2	11.3	(24)
(25)		10%	DB	11.6	12.7	16.7	21.8	26.7	29.8	31.5	32.3	29.9	25.2	18.6	13.4	(25)
(26)		10%	MCWB	8.9	9.2	11.7	15.6	19.5	23.3	25.4	25.5	23.0	18.9	14.4	10.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	13.9	16.9	19.8	23.8	28.0	29.1	29.5	26.9	22.8	18.9	14.6	(27)
(28)		0.4%	MCDWB	17.4	19.4	24.6	27.1	28.2	30.2	31.5	32.2	30.9	26.2	23.3	18.7	(28)
(29)		2%	WB	11.3	12.0	14.8	18.4	22.4	26.1	27.9	28.0	25.9	21.6	17.2	12.9	(29)
(30)		2%	MCDWB	14.5	16.4	21.5	24.5	26.8	29.4	30.7	32.1	30.1	25.7	20.8	16.1	(30)
(31)		5%	WB	10.2	10.6	13.4	17.2	21.4	25.3	27.1	27.3	25.1	20.9	16.0	11.8	(31)
(32)		5%	MCDWB	12.9	13.7	18.5	22.6	25.9	28.9	30.7	31.3	29.2	24.9	19.2	14.8	(32)
(33)		10%	WB	9.1	9.6	12.1	16.3	20.5	24.5	26.3	26.7	24.3	20.0	14.9	10.7	(33)
(34)		10%	MCDWB	11.3	12.1	16.0	20.8	25.1	28.4	30.3	30.8	28.5	24.1	17.9	13.3	(34)
(35)	Mean Daily Temperature Range	MDBR	9.3	9.1	8.9	8.5	7.9	6.7	5.3	6.3	8.2	10.1	9.8	9.3	(35)	
(36)		5% DB	MCDBR	12.6	14.3	16.6	15.4	12.4	8.7	6.8	8.2	9.6	12.3	12.7	12.6	(36)
(37)		5% DB	MCWBR	8.0	8.5	8.1	6.6	4.8	3.7	3.3	4.1	5.2	6.3	7.6	8.3	(37)
(38)		5% WB	MCDBR	11.2	11.7	13.6	11.8	8.9	6.5	5.6	6.8	7.5	9.6	10.1	10.9	(38)
(39)		5% WB	MCWBR	7.7	7.7	7.3	5.7	4.4	3.2	3.1	3.9	4.7	5.8	6.8	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.452	0.509	0.579	0.580	0.591	0.634	0.590	0.566	0.529	0.454	0.411	(40)	
(41)		taud	2.121	1.998	1.878	1.769	1.808	1.813	1.748	1.826	1.849	1.899	2.038	2.113	(41)	
(42)		Ebn at Noon	743	754	749	723	732	723	687	706	692	667	678	701	(42)	
(43)		Edh at Noon	117	150	185	219	214	213	225	204	189	164	127	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.38	3.10	4.13	5.15	6.58	7.39	6.98	6.55	5.35	3.86	2.67	2.11	(44)	
(45)		RadStd	0.16	0.30	0.26	0.56	0.31	0.29	0.38	0.42	0.21	0.19	0.14	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.52	N/A	N/A	+0.97	+0.61	+0.70	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.31	N/A	N/A	+0.57	+0.31	+0.32	N/A	N/A	N/A	N/A

Nomenclature: See separate page